Cal Poly Pomona

College of Business Administration
SINGELYN GRADUATE SCHOOL OF BUSINESS

Mapping Risk & Public Sentiment: Business Insights from Reddit During the Israel-Palestine Conflict

Team members: Cedric Wong, Nhat Le, Yunfeng Fan, Yan Fei (Kerry) Che, Elly Yang, Pratik Kumar

Faculty Co-Leads:

Dr. Mohamed Gomaa

Dr. Nastaran Simarasl

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Agenda

I. Background

i. Introduction
ii. Literature Review
iii. Research Question

II. Analytics

i. Research Methods
ii. Analytic Methods
iii. Findings

III. Recommendations

i. Business Application
ii. Project Limitations

Introduction

- **Problem Statement**

Online discourse during the Israel-Palestine conflict drives brand boycotts and public backlash.

(Chen et al., 2024; Esmaeili & Sobhanifard, 2025)

- **Problem Significance**

Understanding emotions helps companies manage risk and protect brand value.

(Durante & Zhuravskaya, 2018; King, 2011)

Literature Review

Conflict Prediction

- Machine Learning models enhance conflict forecasting (Baron & Rustremi, 2020)
- Conflicts escalate when negative sentiment is anticipated (Durante & Zhuravskaya, 2018)

Corporate Boycott effectiveness depends on:

- Media coverage and target framing (Brayden, 2011)
- Public sympathy in the country of origin (Ding et al., 2020)
- Timing of boycott announcements (Pruitt & Friedmann, 1986)

Research Question

“How do Reddit comments affect the Israel-Palestine conflict and the resulting corporate boycotts?”

Sub-Questions
What sentiment trends emerge in Reddit discussions following major conflict events?
How does online discourse correlate with spikes in boycott-related keywords?
Why do certain brands face stronger backlash during geopolitical crises?

Research Method: Population & Sample

Population:

- Reddit comments about Israel-Palestine.

Sample:

- > 2 million Reddit comments about Israel-Palestine.
- About 58% from r/israelpalestine and r/worldnews.
- Timeframe: 09/02/23 to 01/28/2025.

Data Source:

- Updated daily on Kaggle.com

Research Method: Population & Sample (cont.)

Population:

- Armed Conflict Location & Event Data (ACLED) filtered to Israel-Palestine.

Sample:

- Over 70k armed conflicts between Israel and Palestine.
- Timeframe: 01/01/2016 to 01/24/2025

Data Source:

- Updated daily on ACLEDdata.com

ACLED

Research Method: Population & Sample (cont.)

Population:

- Stocks of Israel-Palestine boycotts companies

Sample:

- McDonald's, Starbucks, and Coca-Cola
- Daily prices
- Timeframe: 10/01/2023 to 01/24/2025

Data Source:

- Stock Exchange Websites (NYSE, NASDAQ).

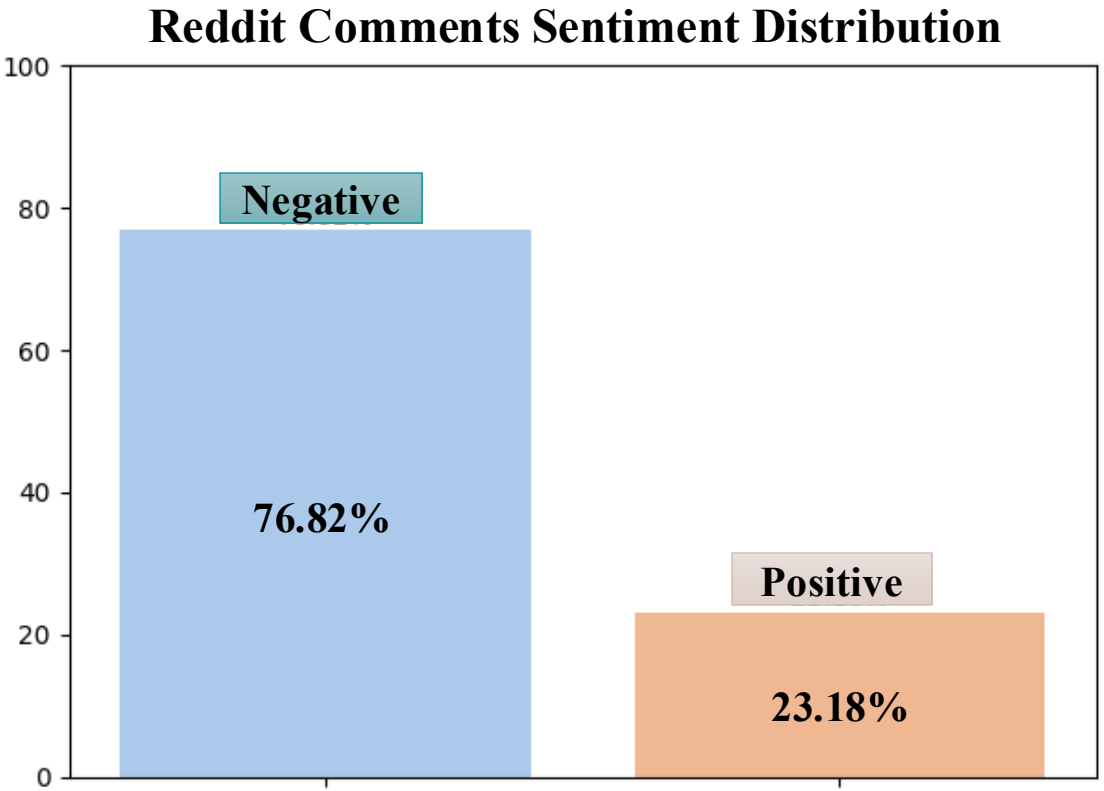
Research Methods

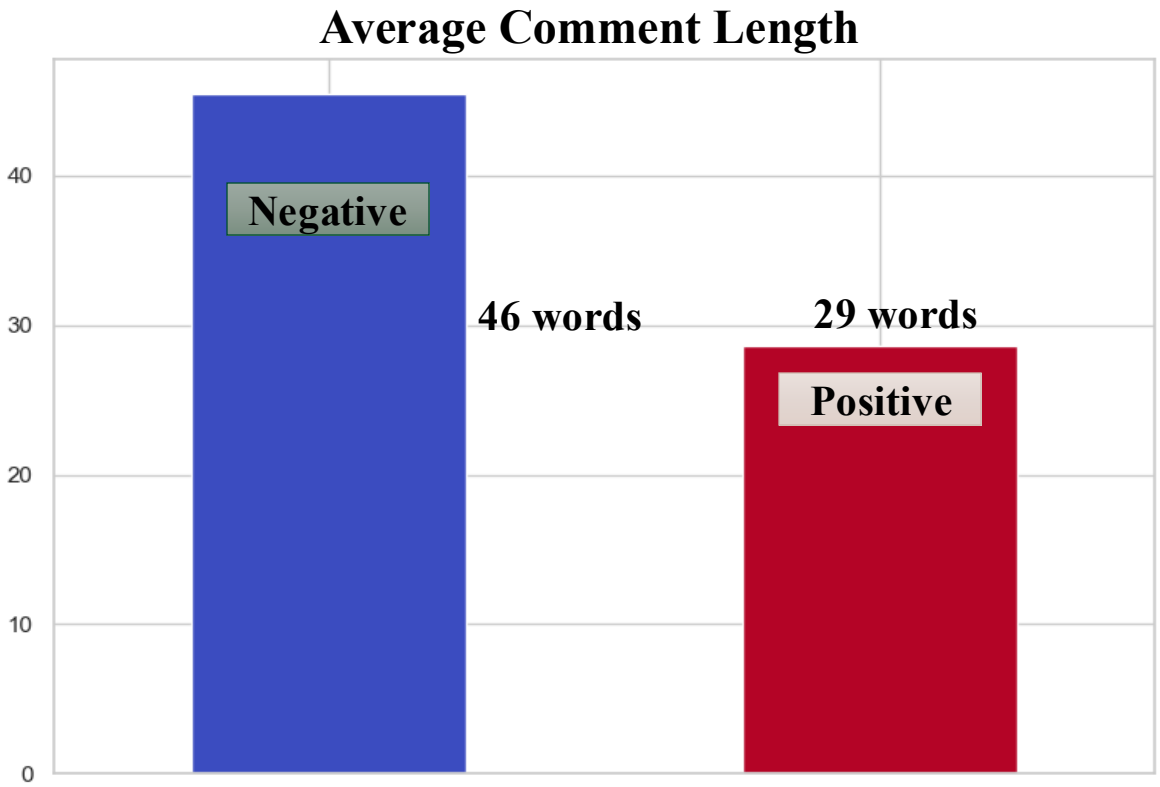
Analytic Objective (AO)	Research Methods
1. Sentiment Analysis	Identify emotional tone using AI-based (BERT) sentiment tools.
2. Demographic Modeling	Visualized conflict events by location and time using ACLED data
3. Text Attribute Impact	Features like comment complexity and wording affected engagement.
4. Temporal Trends	Tracked sentiment alongside stock price shifts during boycotts.
5. Conflict Categorization	Categorized conflict into types (protests or battles) to analyze pattern.
6. Conflict Prediction	Predicted conflict escalation using machine learning on ACLED data.

AO1 - Sentiment Skews Strongly Negative

- 77% comments are negative

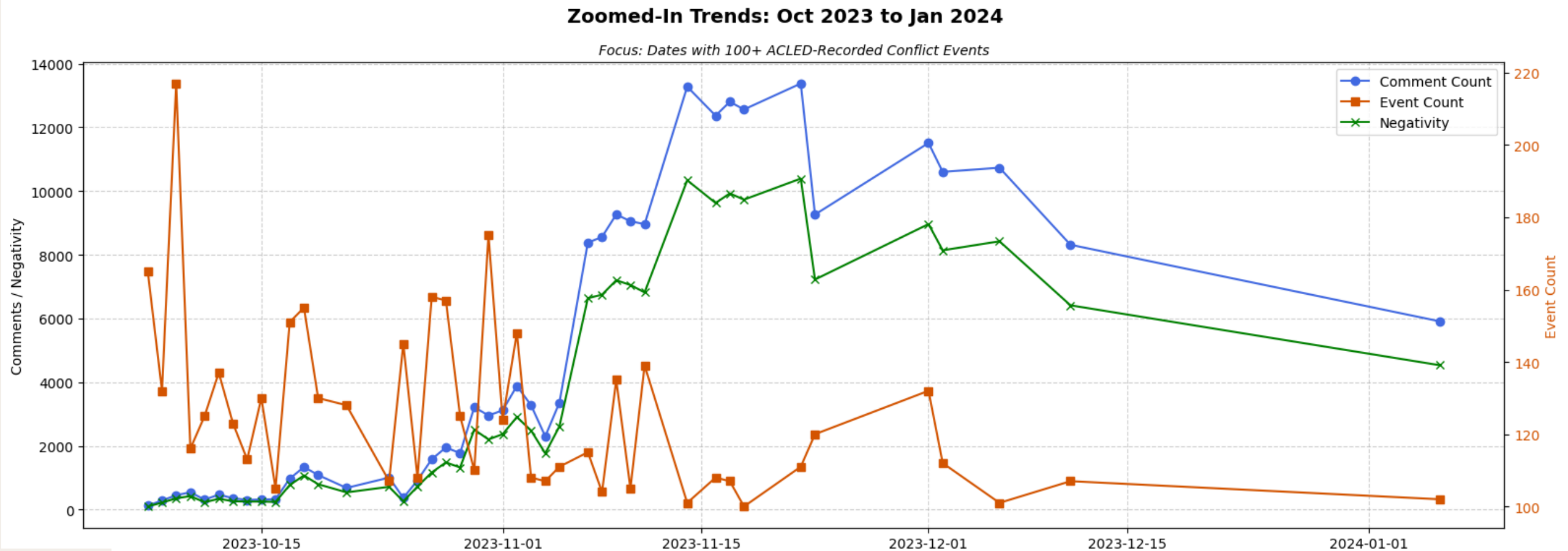
- Negative comments are 60% longer
- Longer Comments = deeper emotions





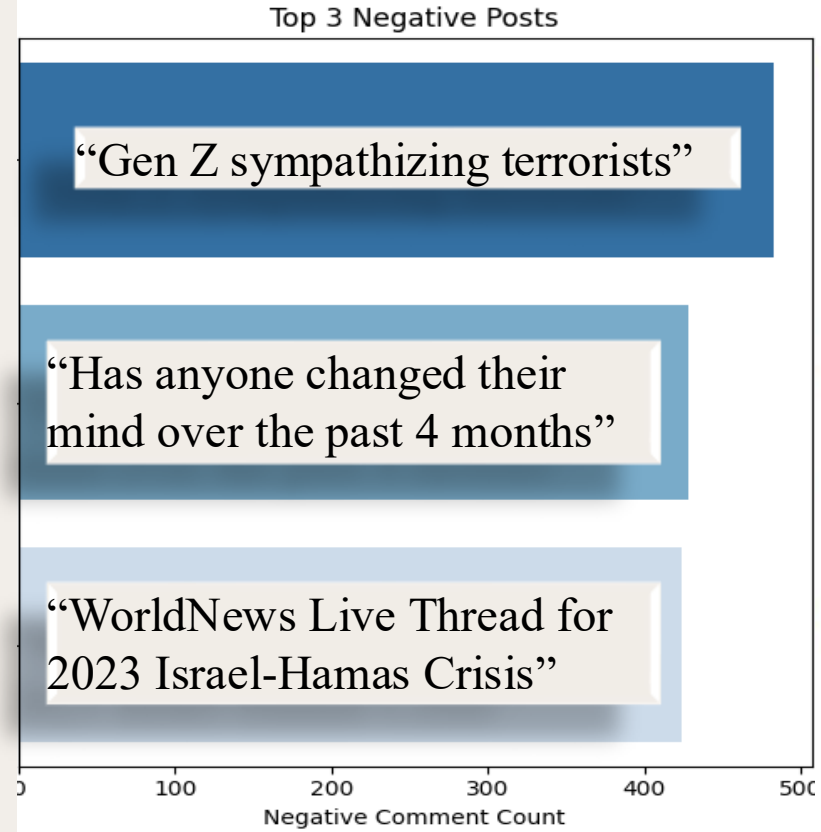
AO1 - Conflict Drives Sudden Negativity Spikes

- Conflict days trigger negativity surges
- Spikes last few days



AO1 - Negative Sentiment Is Precision-Directed

- Top posts blame targeted groups
- Anger spreads through emotionally charged posts



- Word Cloud of Negative Comments:

  "terrorist," "crime," dominate.

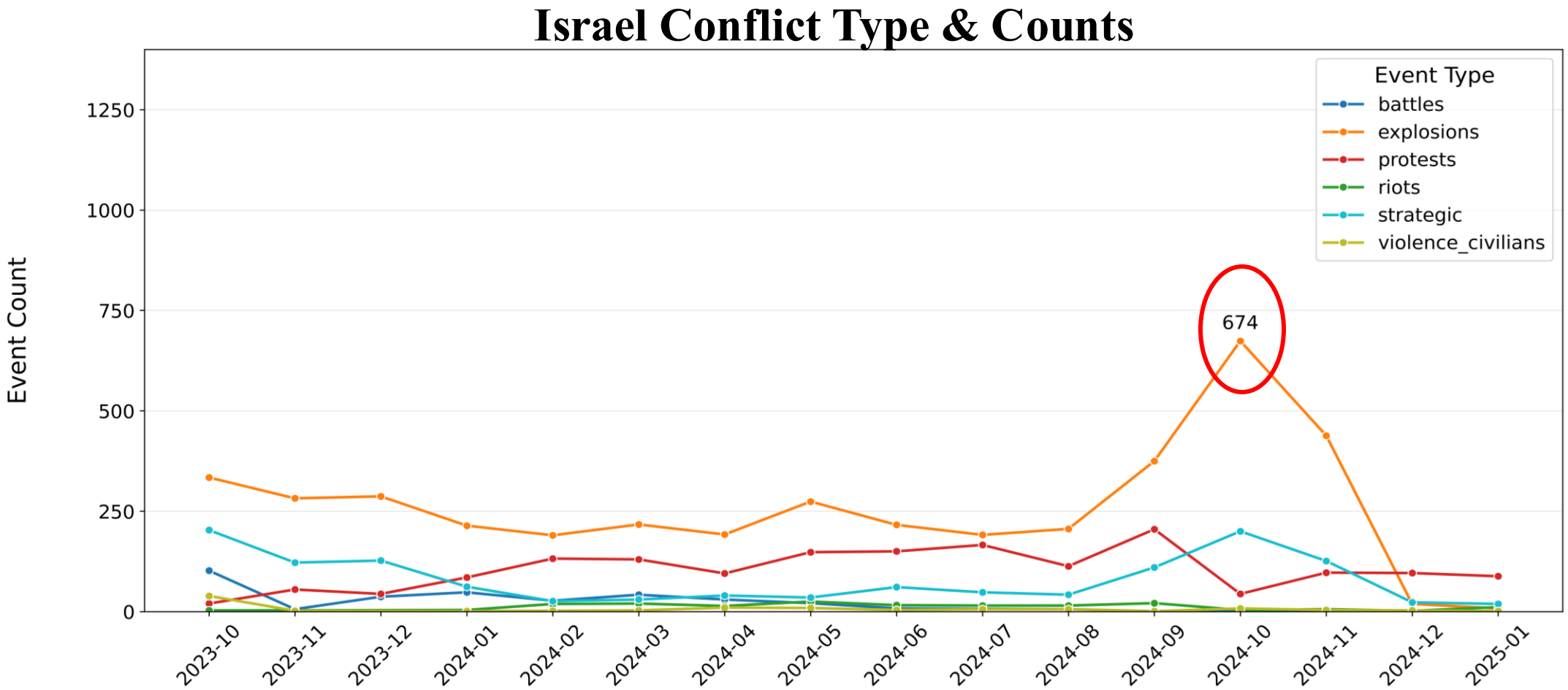
west bank
talk
mean
come
reason
sure
yes
work
time
war
crime
state solution
middle east
lol
place
example
bad
well
jewish people
especially
year ago
literally
instead
kind
cause
issue
problem
matter
fact
guess
course
consider
case
believe
probably
wrong
etcfuck
exactly
hope
claim
international law
support hama
russia
ethnic cleansing
pro israel
imagine
leave
human shield
feel
american
expect
israelforce
obviously
agree
likely
basically
lot
understand
remember
true
pro palestinian
definitely
good
little
idea
make
question
find
simply
today
wonder
iran
speak
human right
long
care
change
possible
day
guy
mention
sound
ukraine
view
show
end
clear
hear
happen
hand
support israel
attack israel
unfortunately
absolutely
palestinian state
israeli government
protest
assume
call
help
include
clearly
jewish state
conflict
explain
palestinian people
actual
stupid
use
amp
sort
man
forget
plan
second
word
wait
real
america
bring
year old
nation
state
social media
turn
one
stand
side
difference
decade
certainly
country
involve
zionist
civilian
kill civilian
wish
okay
stuff
answer
simply
call
start war
woman child
start
suppose
ignore
person
position
lie
honestly
fair
despite
line
compare
arab country
follow
away
war israel
friend
load
nation
suggest
response
close
israel israel
anymore
argue
far
different
dude
big
similar
opinion
people live
kill people
terrorist organization
hell
run
russian
kill civilian
civilian casualty
israel palestine
palestinian people
netanyahu
order
decade
love
involve
sionist
israel right
certainly
country
side
show
view
ukraine
shit
issue
mention
sound
great
absolutely
support israel
attack israel
unfortunately
palestinian state
israeli government
protest
assume
call
start war
woman child
start
include
clearly
jewish state
help
possible
care
word
second
plan
change
guy
day
conflict
explain
palestinian people
actual
stupid
use
amp
sort
man
forget
wonder
iran
speak
human right
long
today
question
make
idea
pro palestinian
simply
find
good
little
definitely
exactly
hope
claim
international law
support hama
russia
ethnic cleansing
pro israel
imagine
leave
year ago
especially
human shield
feel
american
expect
israelforce
obviously
agree
likely
basically
lot
understand
remember
true
place
example
bad
well
jewish people

AO2. Geographic Modeling of Conflict & Sentiment

- **Objective:**
  - Explore how violent conflict events in Israel and Palestine influence public sentiment on Reddit.
- **Methodology**
  - Merge ACLED conflict data with Reddit sentiment
  - Tag comments by region using keywords
  - Use BERT model to compute daily sentiment scores
  - Apply lagged regression (t+1 to t+3) to analyze delayed emotional response

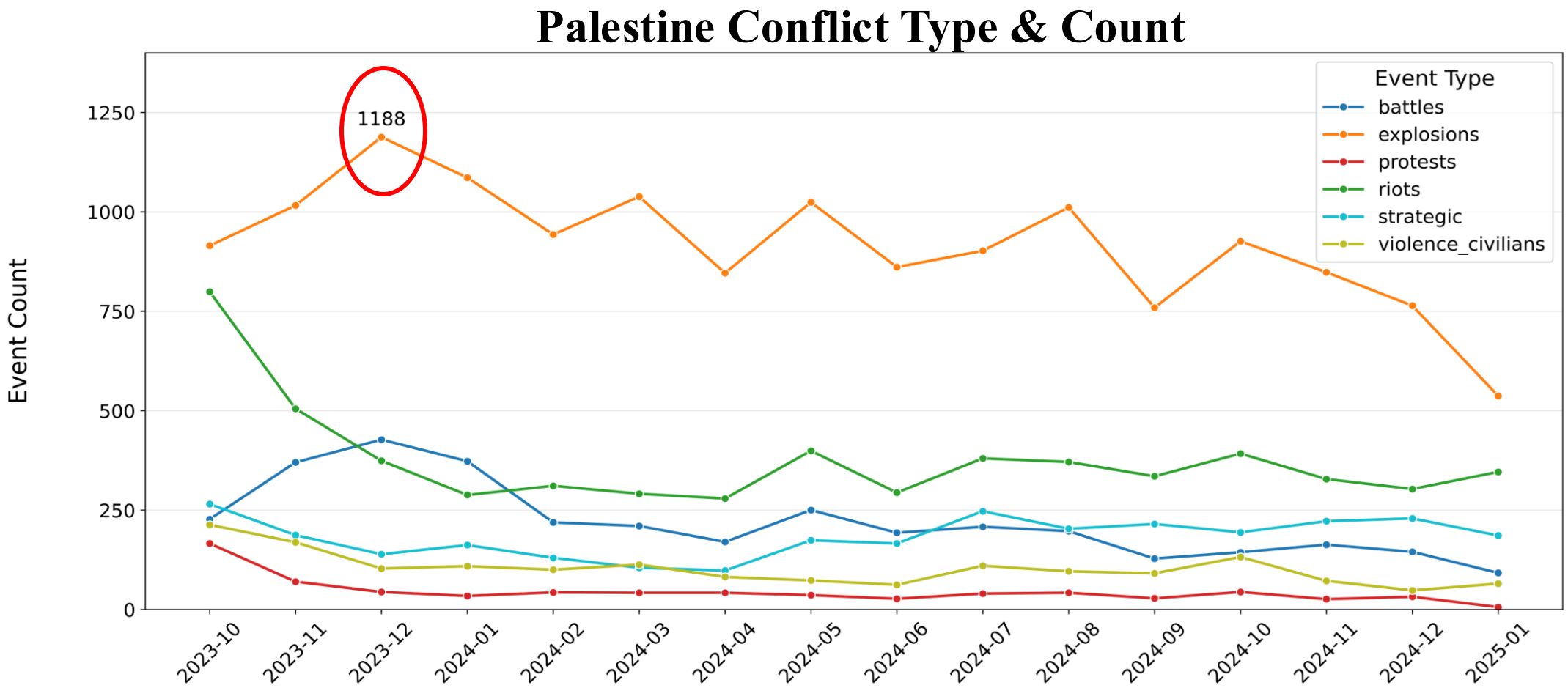
AO2 - Conflict Event Type (Israel vs. Palestine)

- **<u>Country Comparison:</u>** Israel showed sharper spikes, especially in explosions.



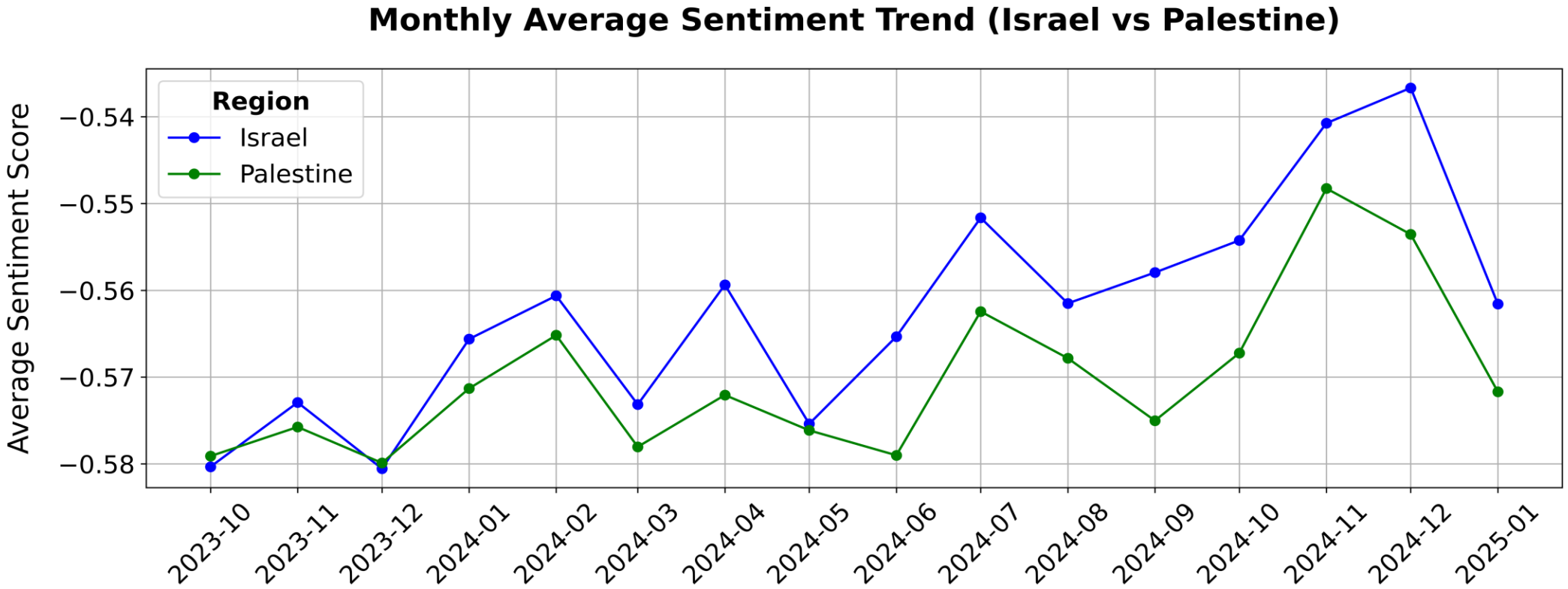
AO2 - Conflict Event Type (Israel vs. Palestine)

- **<u>Country Comparison</u>:** Palestine experienced a higher total number of events.



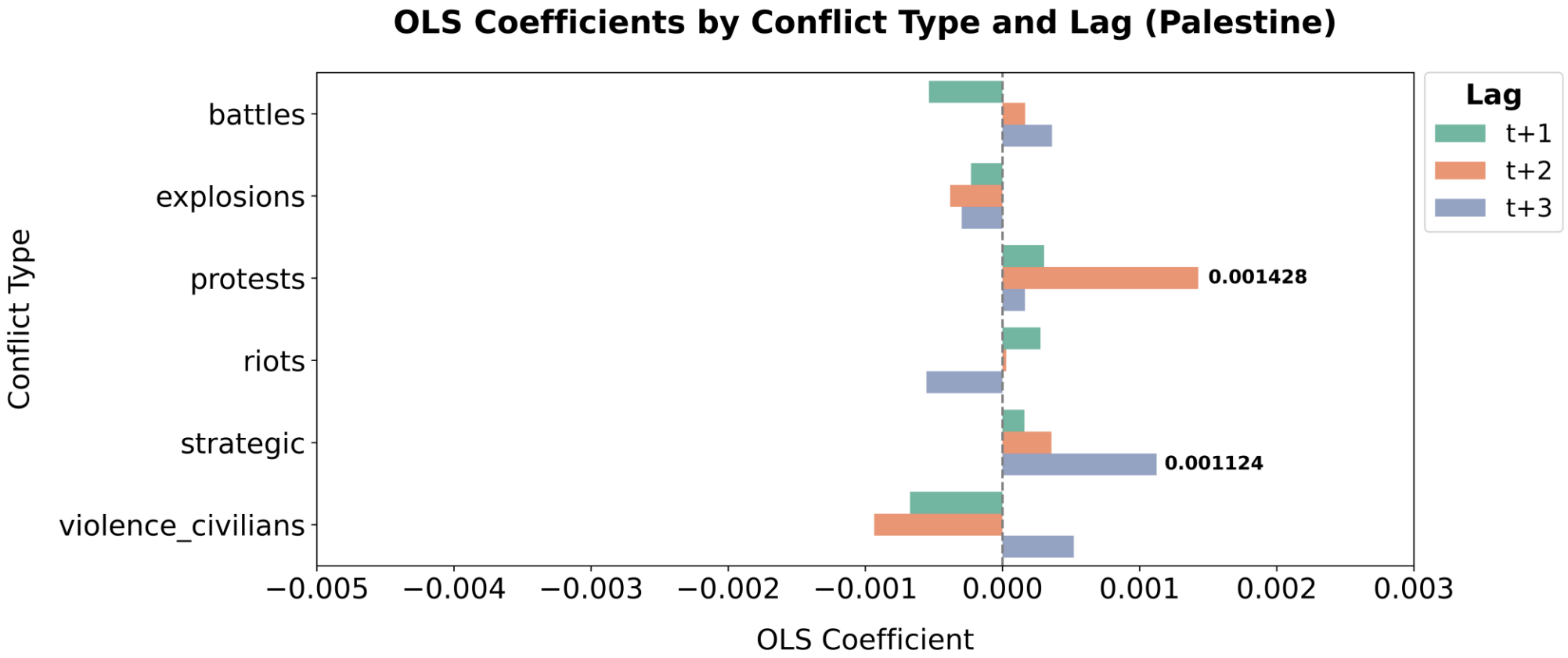
AO2 - Sentiment Trends

- Sentiment Analysis Setup:
  - Region-specific sentiment derived from BERT model (daily average)
  - Comments filtered for Israel/Palestine via keyword tags
- Trends Observed:
  - Both negative score but steady increase over time
  - Palestine receives more negative sentiment score



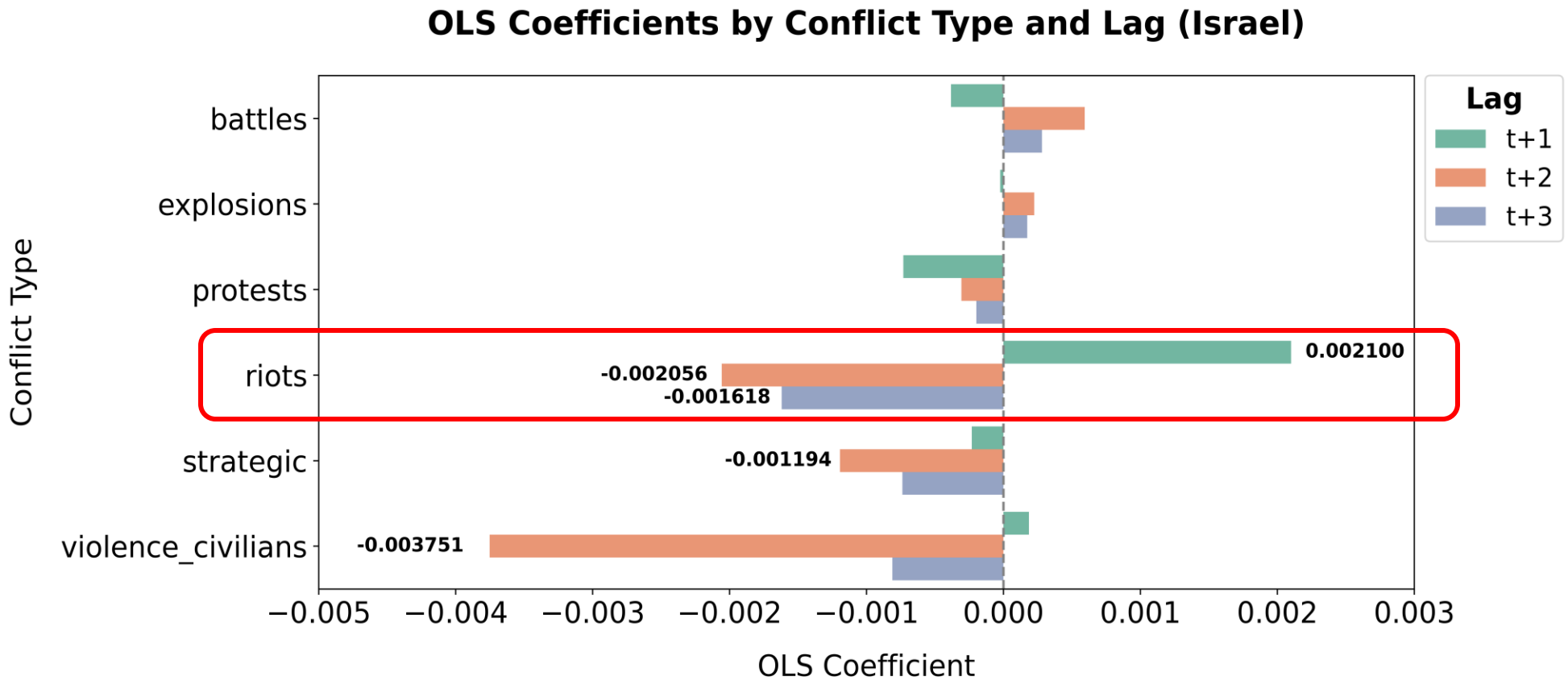
AO2 - Regression: Conflict Type vs. Sentiment Impact

- **Palestine:**
  - Protests → positive sentiment (t+2)
  - Strategic → Positive effect(t+3)
  - Riots → negative effect (t+3)
  - Violence target civilians → change rapidly



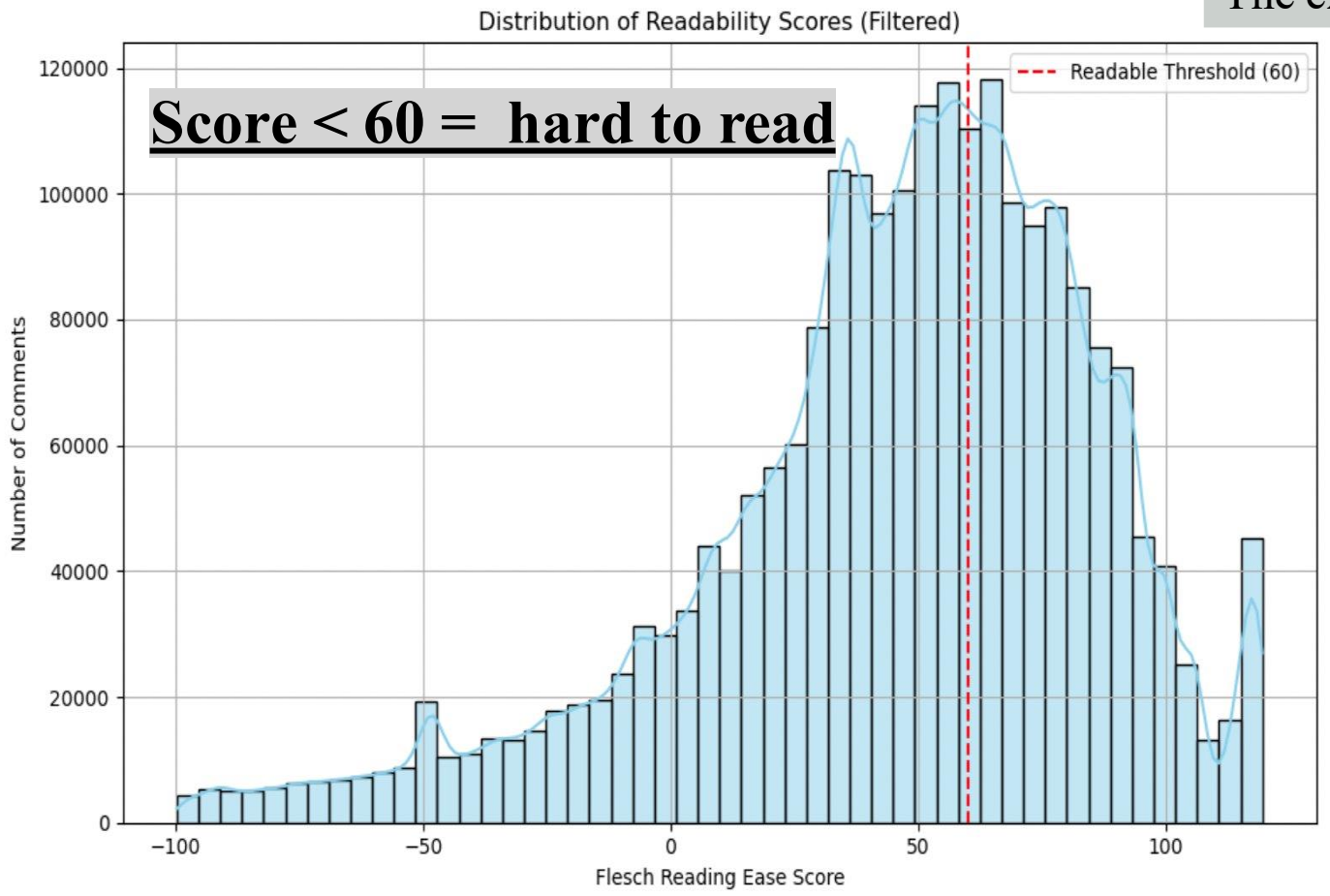
AO2 - Regression: Conflict Type vs. Sentiment Impact

- **Israel:**
  - Violence Against Civilians → strong negative impact (t+2)
  - Riots → strong positive(t+1) rapidly decrease for (t+2) and (t+3)



AO3. Text Readability

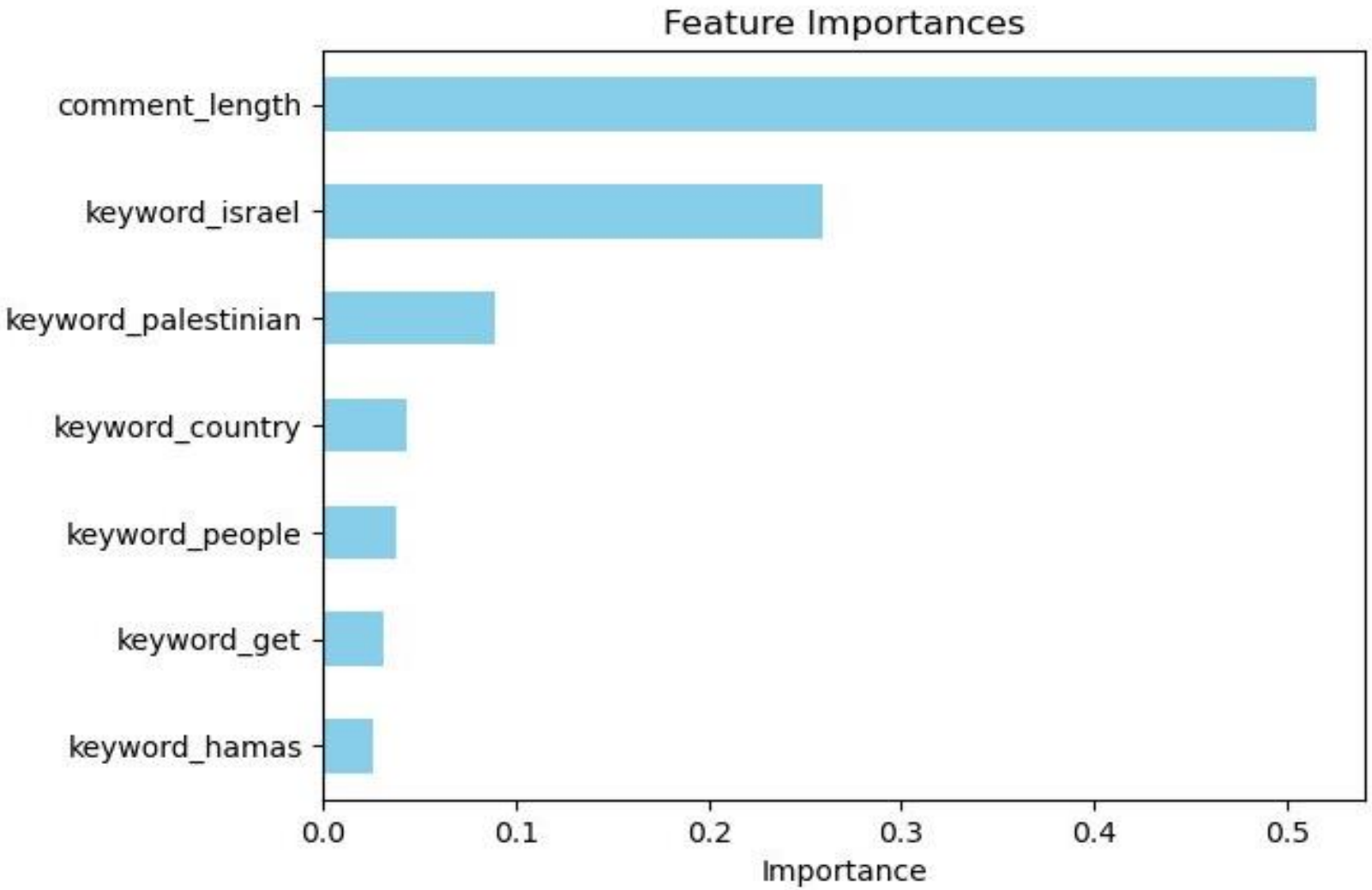
Reddit Comment	Score
Their voices are absolutely beautiful.	15.64
I was meaning the historical reasons…	30.20
Ya got brainwashed pardner	75.88
Are Palestinians still living in Jordan…	40.69
The excuse that the Palestinians have a valid…	35.62



- Reddit language is noisy (sarcasm, slang & informal phrasing)
- Most comments has low-readability

AO3. Text Feature Importance

- Common words used: “Israel,” “People,” “War”, "Palestinian“
- Keyword “Israel” boosts visibility
- Comment length is the strongest predictor



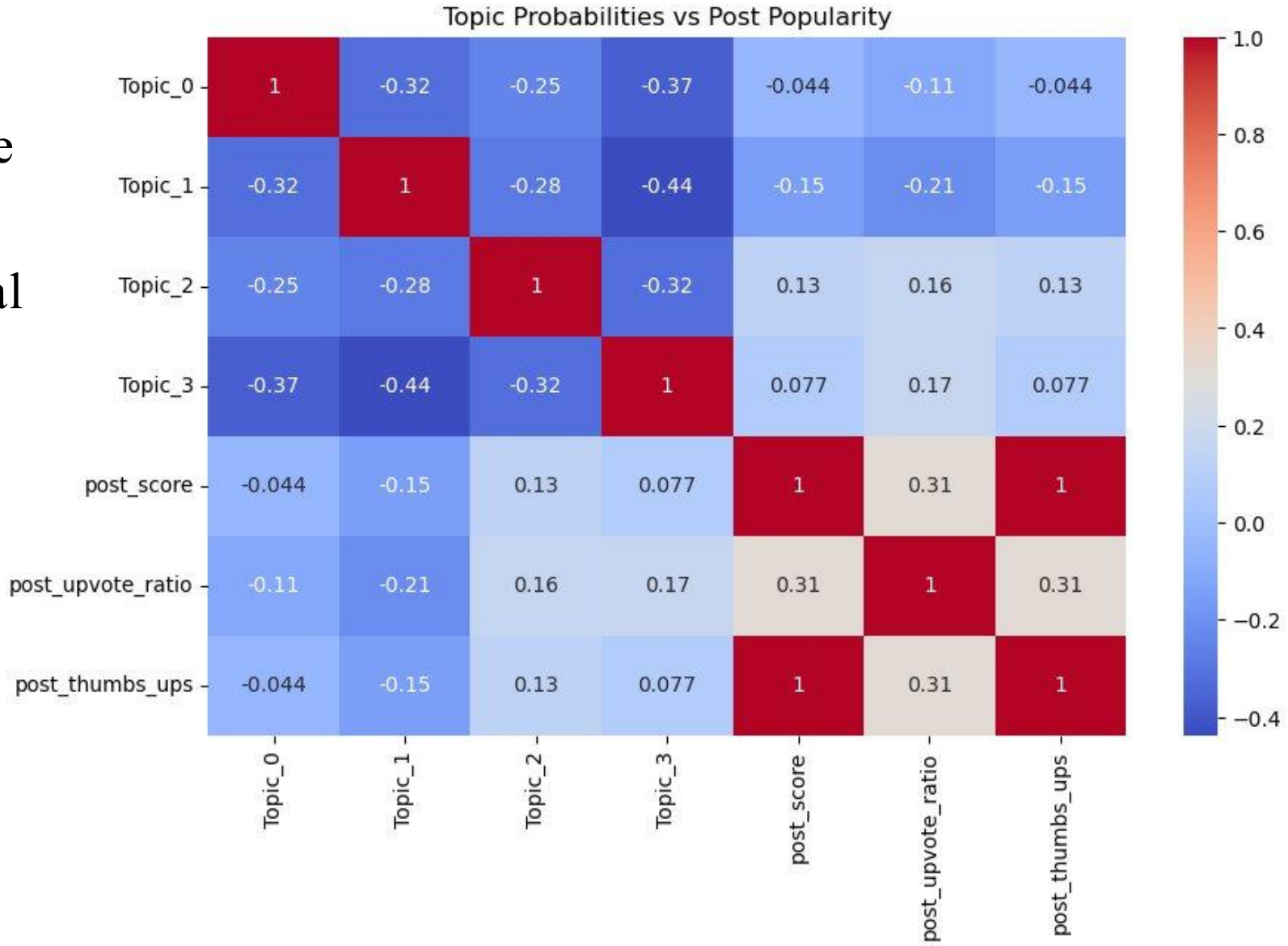
AO3. Topic Modeling Themes

- Topic 0 and 1, center conflict-heavy themes (e.g., :"Hamas", "Kill")
- Topic 2 and 3, neutral/general tone themes

Topic	Theme
0	*“Hamas”, “Israel”, “Gaza”, “kill”, “civilian”, “people”*
1	*“Israel”, “Palestine”. “jew”, “people”, “Palestine”, “State”*
2	*“us”, “Israel” , “would”, “war”, “country”, “get”*
3	*“like”, “people”, “get”, “say”, “see”, “know”*

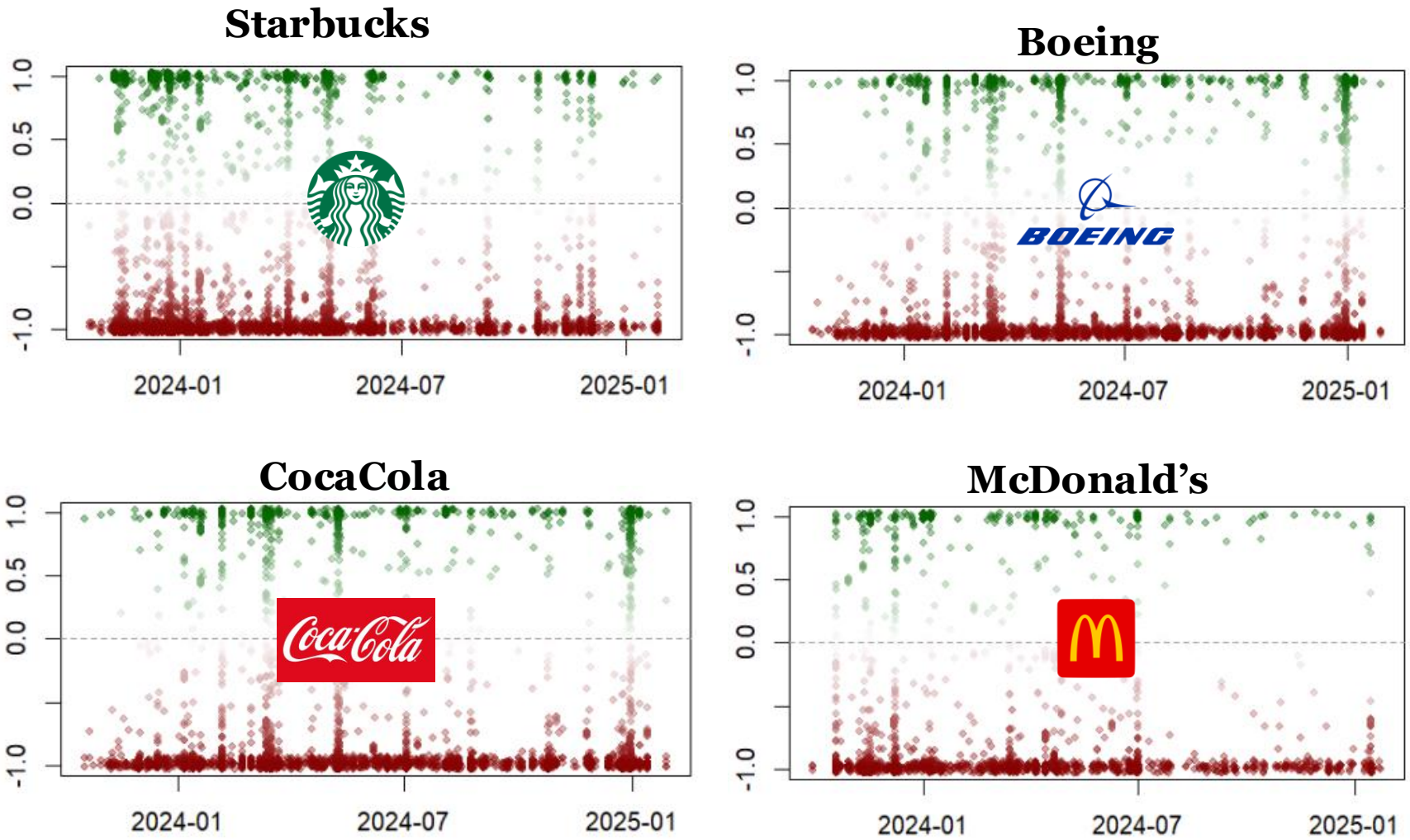
AO3. Correlation of User Engagement

- Conflict and Identity-focused comments receive lower engagement
- Neutral and conversational topics perform better
- Less emotional, more general content draws more upvotes



AO4. Temporal Trends

- Sentiment do not move stocks
- Stock movements likely had other causes
- No clear trend across time
- Results were weak for all brands
- Sentiment tracks brand image, not price
- Use news and data together



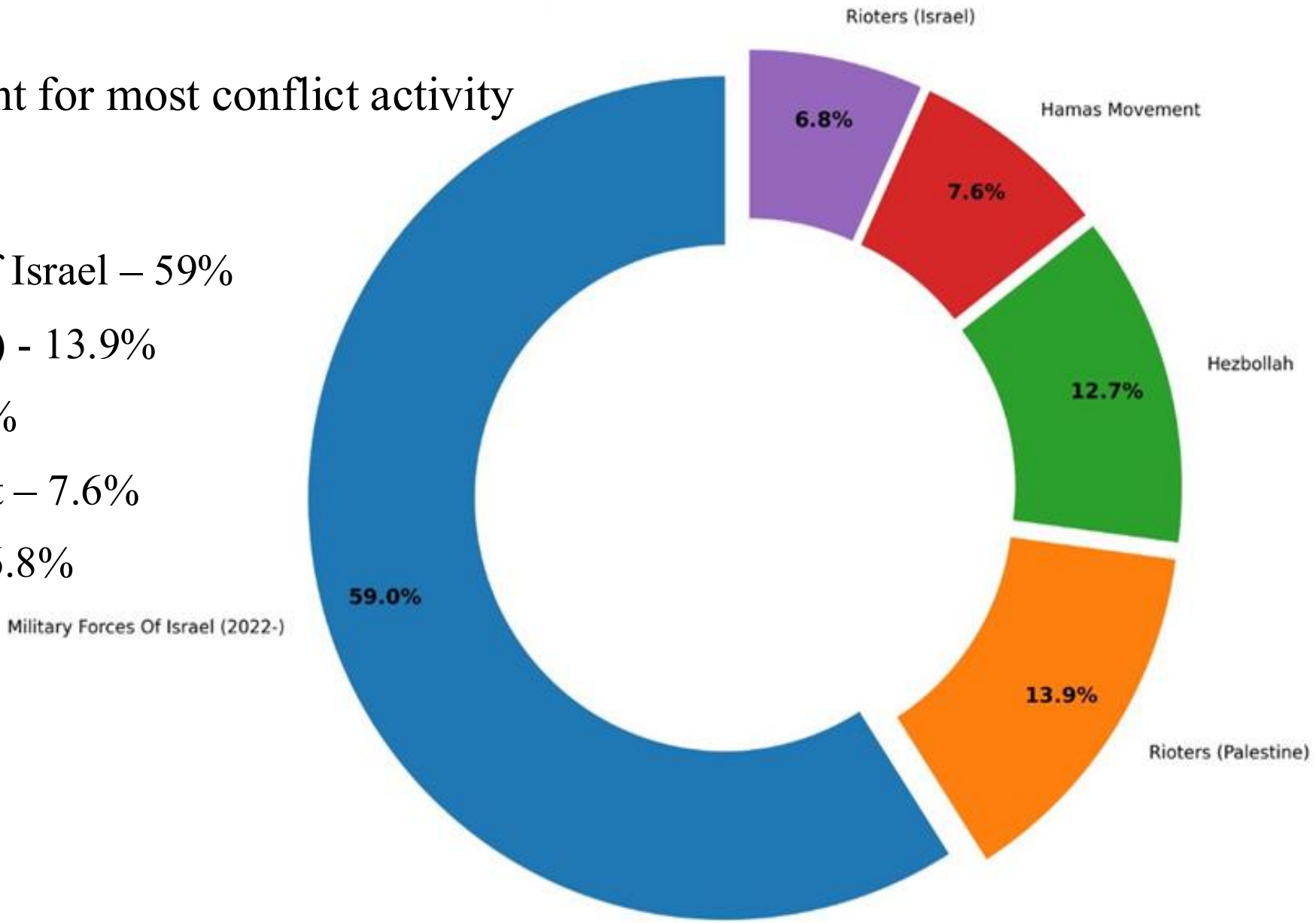
(y-axis: Sentiment Score)

AO5. Conflict Categorization

- Israeli forces account for most conflict activity

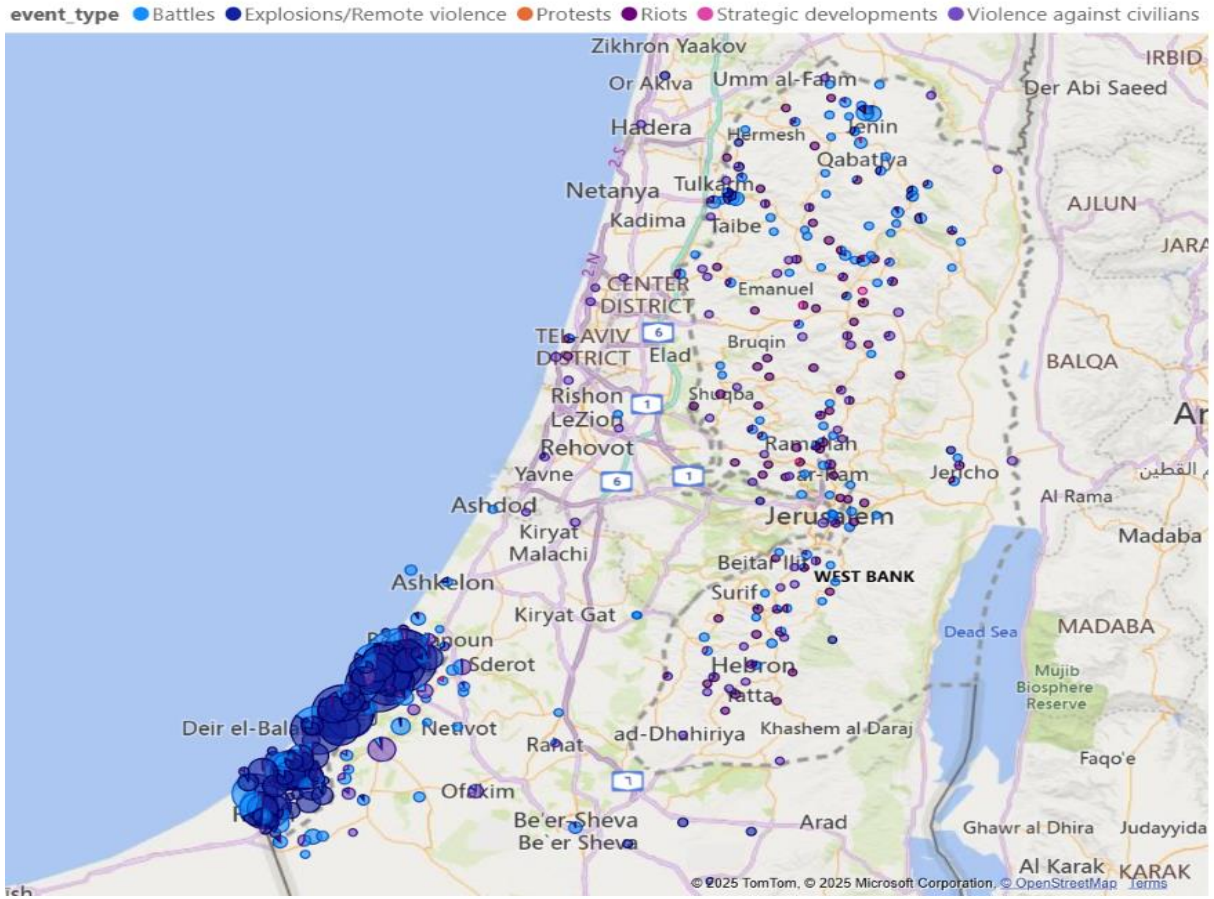
Top 5 Actors

1. Military Forces of Israel – 59%
2. Rioters (Palestine) - 13.9%
3. Hezbollah – 12.7%
4. Hamas Movement – 7.6%
5. Rioters (Israel) - 6.8%

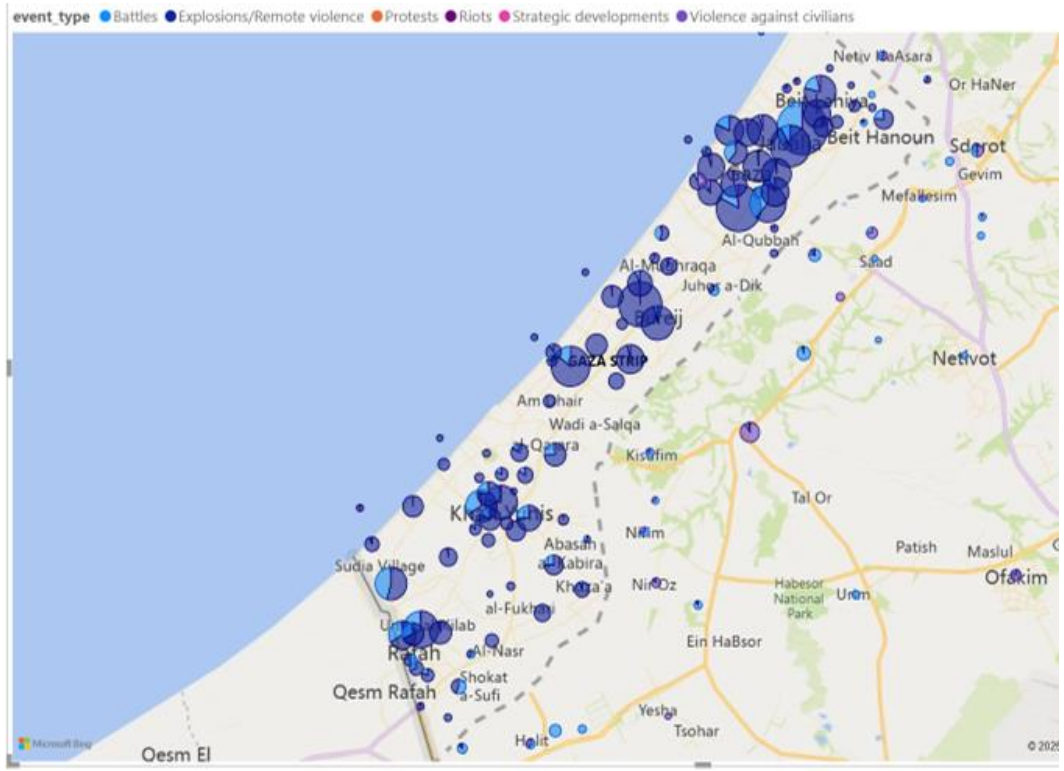


AO5. Conflict Categorization- Geographic Hotspots

- Gaza region shows intense conflict
- Remote violence dominates hotspot regions
- Conflict clusters reveal geographic patterns



Conflict distribution across Israel-Palestine

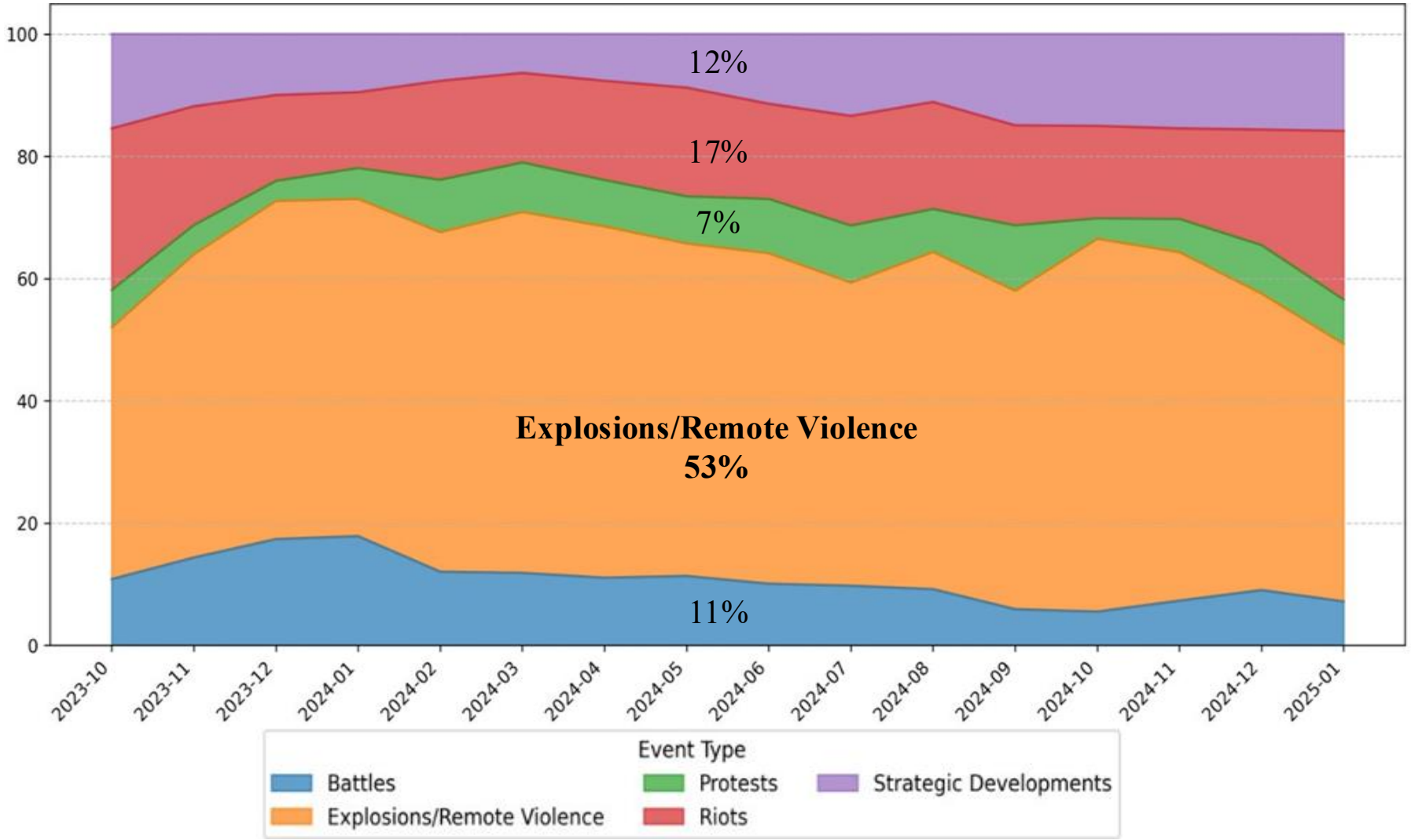


Zoom on Gaza: dense conflict clusters

- Gaza requires targeted monitoring efforts

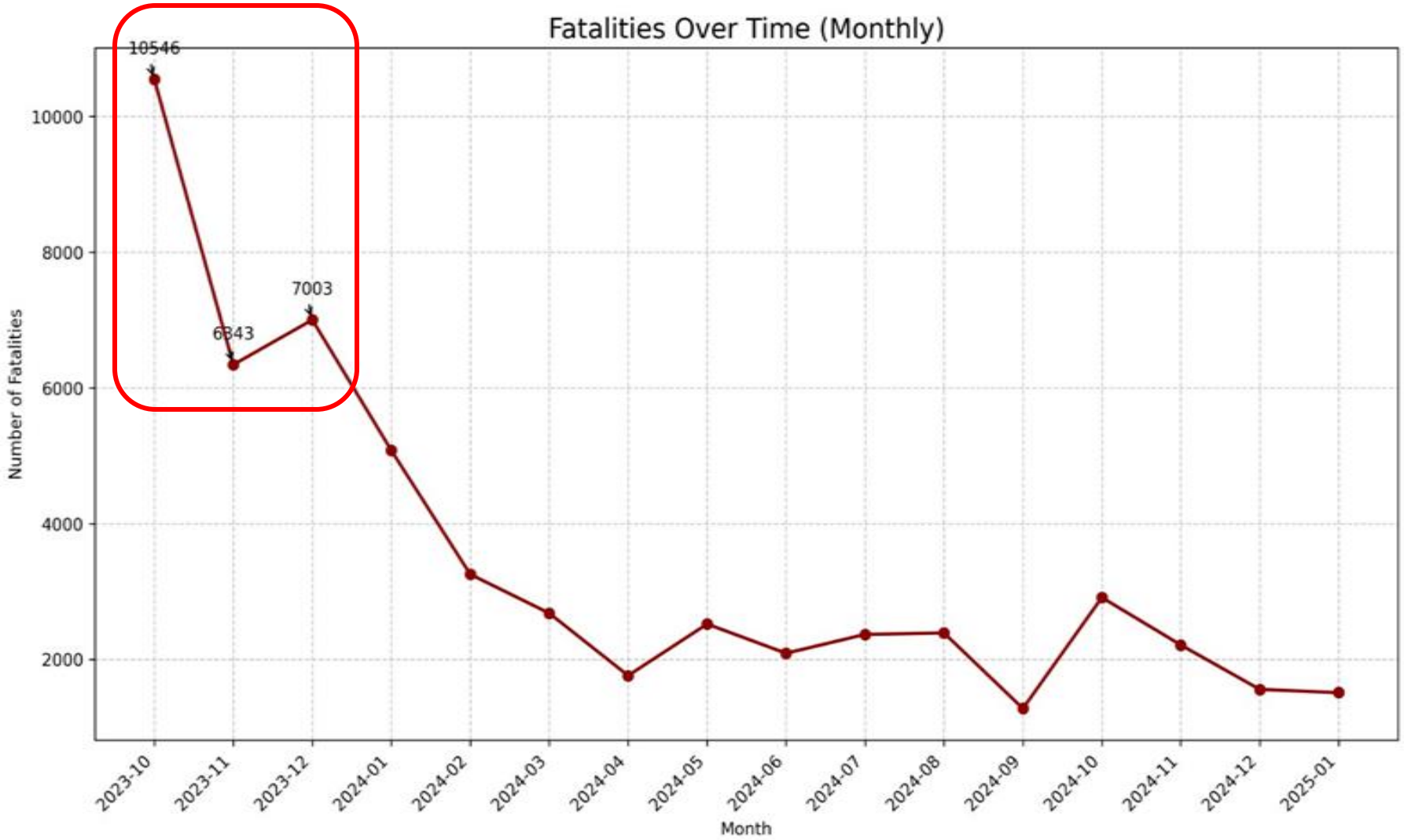
AO5. Conflict Categorization

- Conflict events peaked sharply in October 2023, driven by remote violence



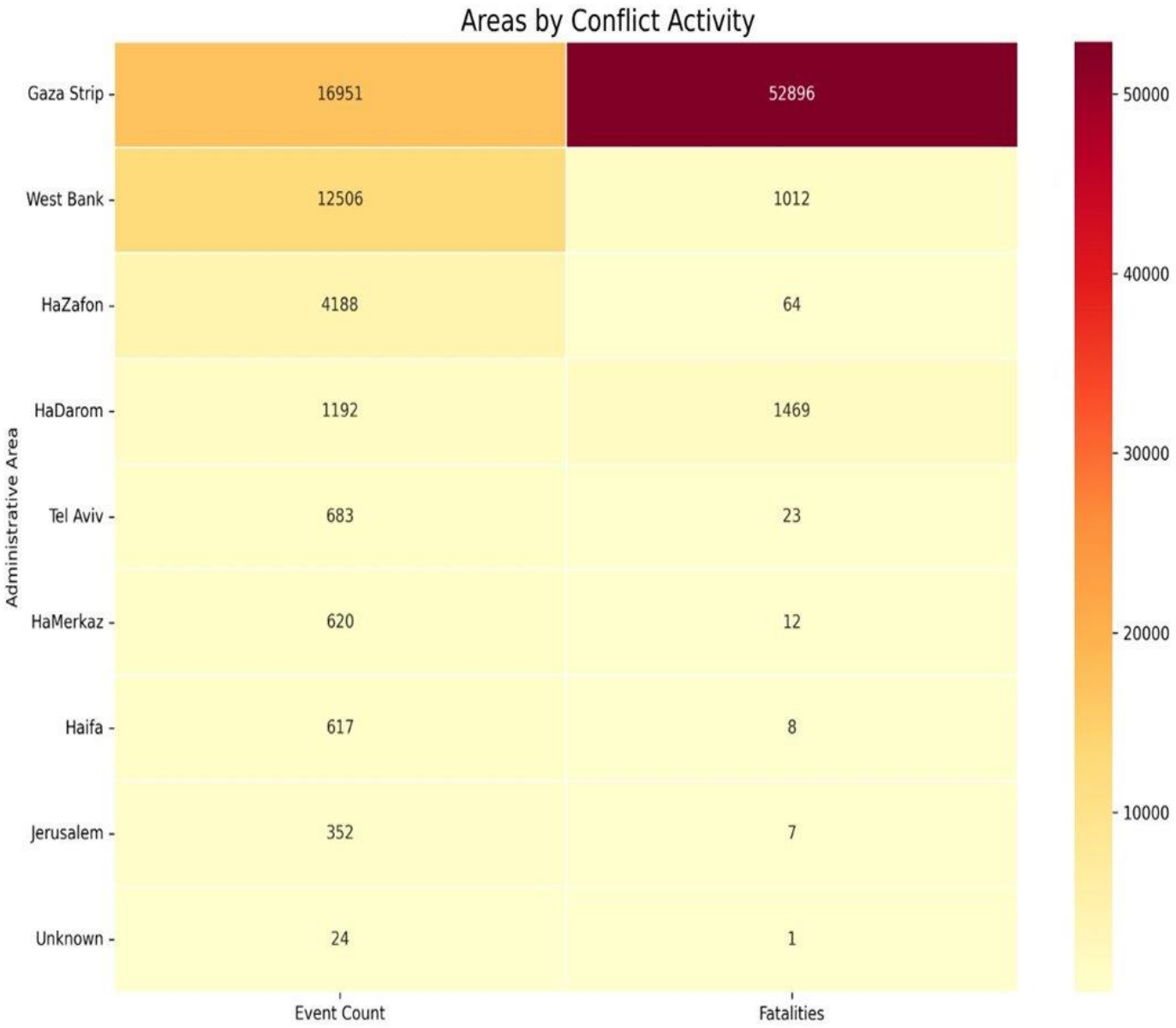
AO5. Conflict Categorization

- Monthly fatalities exceeded 10,000, signaling crisis escalation



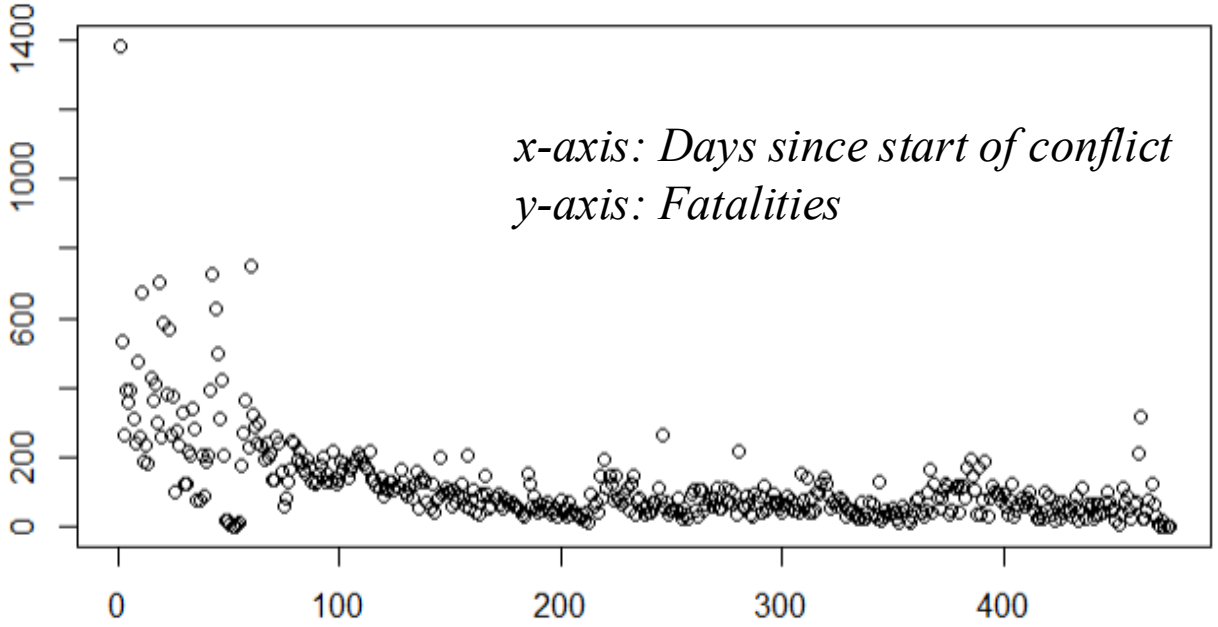
AO5. Conflict Categorization

- Gaza Strip leads all regions in both events and fatalities
- West Bank ranks second in event count but much lower in fatalities
- Southern districts show high fatality-to-event ratios



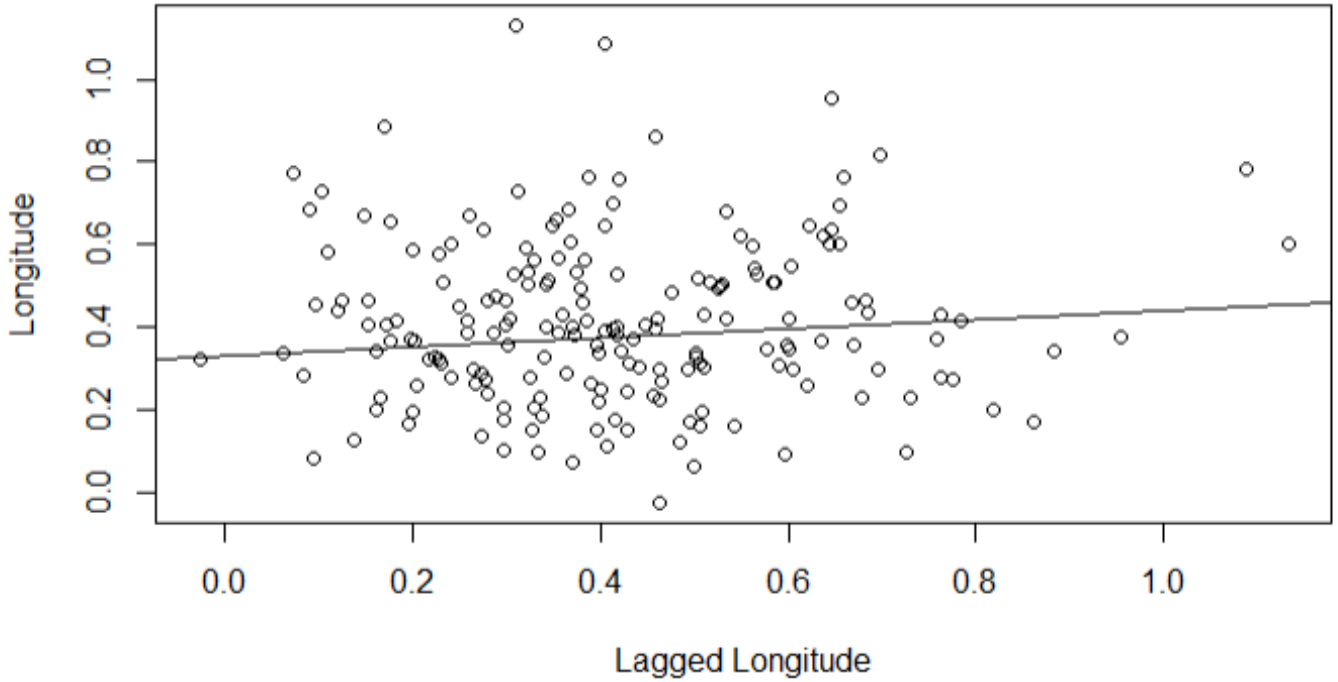
AO6. Conflict Predictive Analysis

- Fatality count trends down over time
- Average of 117 fatalities per day
- Conflict location averages out to Beit Elazari, Israel



AO6. Conflict Prediction– Linear Regression

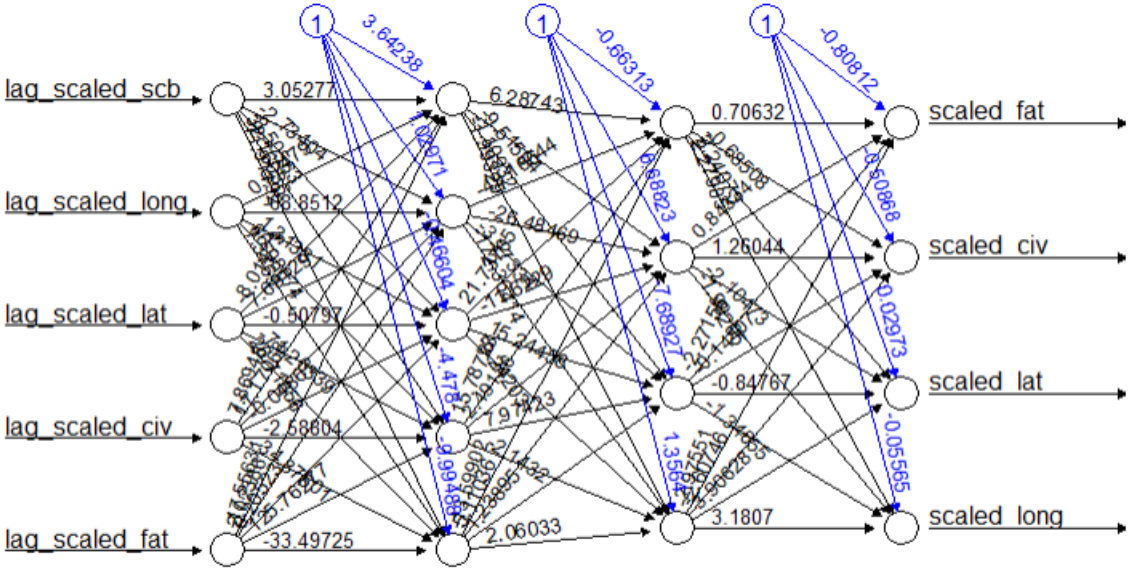
- Weak correlation within variables when lagged
- No individual coefficient demonstrated significance
- Intercepts were all highly significant
  - Expected value more predictive than any independent variables



AO6. Conflict Prediction– Neural Network

- Compared models using Mean Absolute Percentage Error
- Both models performed similarly
  - Fatalities and percentage of civilian targeted attacks too volatile to predict
  - Location predictions performed better under linear regression
  - Center of attacks narrowed down to an 8.62-mile radius on average

	Latitude	Longitude	Civilian Targeting	Fatalities
Regression MAPE	0.34%	0.22%	23.82%	234.68%
Neural Network MAPE	0.38%	0.24%	25.61%	228.79%



Findings Summary

Analytic Objective	Finding(s)
AO1. Explore how people feel about the conflict by using sentiment analysis on Reddit comments.	• Conflict days spike negative sentiment • Civilian harm draws stronger reactions • Specific details boost public attention
AO2. Explore how violent conflict events in Israel and Palestine influence public sentiment on Reddit.	• Civilian-targeted violence triggers strong negative sentiment • Riot changes sentiment rapidly • Sentiment impact varies more by conflict type
AO3. Understand how text features of Reddit comments influence user engagement.	• Longer comments do not boost engagement • Clearer language improves visibility • Conflict keyword "Israel" increase reach • Emotional heavy content did not drive attentions

Findings Summary

Analytic Objective	Finding(s)
AO4. Finding the link between Reddit sentiment and corporate stock performance during the conflict.	• Sentiment & stock have weak correlation • Monitor sentiment for brand awareness • Incorporate traditional metrics for better results
AO5. Understand conflict patterns through event classification and regional mapping using ACLED data.	• Remote violence is the most frequent event type • Israeli forces lead in conflict involvement • Gaza Strip shows highest conflict and fatality levels
AO6. Predict conflict statistics using sentiment data in multiple statistical models.	• Fatality count and proportion of civilian targeted attacks too volatile to be predicted • Location of attack can be predicted to an 8-mile radius • Choice of model has little effect on predictive power

Recommendations (1)

Real-Time Conflict Monitoring

- Track sentiment shifts across regions
- Detect spikes in public attention
- Identify risks to brand reputation
- Respond faster to emerging crises

→ **Helps protect PR and market position**

Recommendations (2)

Predictive Early Alert Systems

- Forecast violence zones in advance
- Tag high-risk events for alerts
- Support timely risk mitigation planning
- Guide resource allocation for safety

→ **Enables proactive crisis management**

Recommendations (3)

Integrate News & Financial Intelligence

- Merge conflict + market news signals
- Understand stock trends after crises
- Contextualize social sentiment impacts
- Avoid overreacting to short-term noise

→ **Informs smarter investment decisions**

Recommendations (4)

Brand Messaging Strategy

- Use clear, neutral communication
- Avoid emotional or polarizing language
- Strengthen trust during global crises
- Tailor tone by region and audience

→ **Builds long-term brand loyalty**

Project Limitations

- Conflict-Specific Scope: Focused only on Israel-Palestine conflict
- Platform Bias: Based solely on Reddit content
- Language Scope: English comments only
- Context Gaps: No user identity or intent analysis
- Time Sensitivity: Reflects a specific conflict period

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Q&A

Cedric Wong: cedricnwong@cpp.edu

Nhat Le: ndle@cpp.edu

Elly Yang: ellyyang@cpp.edu

Yan Fei (Kerry) Che: kerryche@cpp.edu

Yunfeng Fan (Franklin): yunfengfan@cpp.edu

Pratik Kumar: pratikkumar@cpp.edu

Thank you!